

- 7 The curve C has equation $y = f(x)$, where $f(x) = \frac{x^2 + 2}{x^2 - x - 2}$.

(a) Find the equations of the asymptotes of C . [2]

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(b) Find the coordinates of any stationary points on C , giving your answers correct to 1 decimal place. [4]

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- (c) Sketch C , stating the coordinates of any intersections with the axes.

[3]

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- (d) Sketch the curve with equation $y = \frac{1}{f(x)}$.

[2]

- (e) Find the set of values for which $\frac{1}{f(x)} < f(x)$. [4]

[illegible]